

財政學 HW2

1. (112考古) Assume there are two sectors in the economic system, the corporate sector and the non-corporate sector. The marginal value product of capital in the corporate sector is $VMP_c = 150 - K_c$, that K_c is the amount of capital used in the corporate sector. The marginal value product of capital in the non-corporate sector is $VMP_n = 130 - 3K_n$, that K_n is the amount of capital used in the non-corporate sector. The total amount of capital in the society is 100 units. Explain with a diagram, in the case without taxation, what are the amounts of capital in the corporate and non-corporate sectors respectively? Furthermore, assume that the government imposes a tax of 12 units per unit of capital on the corporate sector. What are the amounts of capital and the excess burden of taxation in each sector after tax? Please analyze it with graphs, too.

2. (Rosen 14.3) For commodity X , average cost is equal to marginal cost at every level of output. Assuming that the market for X is competitive and the demand curve is linear, analyze the effects when a unit tax of u dollars is imposed. Now analyze the effects of the same tax assuming that the market for X is a monopoly. Discuss the differences.

3. (Rosen 14.5) In an effort to reduce alcohol consumption, the government is considering a \$1 tax on each gallon of liquor sold (the tax is levied on producers). Suppose that the demand curve is $Q^D = 500,000 - 20,000P$ (where Q^D is the number of gallons of liquor demanded and P is the price per gallon), and the supply curve for liquor is $Q^S = 30,000P$ (where Q^S is the number of gallons supplied).
 - a. Compute how the tax affects the price paid by consumers and the price received by producers.
 - b. How much revenue does the tax raise for the government? How much of the revenue comes from consumers, and how much from producers?

c. Suppose that the demand for liquor is more elastic for younger drinkers than for older drinkers. Will the liquor tax be more, less, or equally effective at reducing liquor consumption among young drinkers? Explain.

4. (Rosen14.1) In 2009, it was proposed that the state of Nevada create an entertainment tax that “would require the state’s 25 legal brothels to give the state some money on a per-transaction basis” [Friess, 2009]. Discuss the likely incidence of such a tax. Use an appropriate diagram as the basis for your discussion.

5. 請從《家庭收支調查報告》的統計數據（如吉尼係數、所得五等分位差距倍數），說明我國近年的所得分配趨勢。